

PROFESSIONAL FULL-STACK ENGINEERING ROADMAP

Engineering Workflow & Git Collaboration (Company-Grade)

A structured reference covering professional Git collaboration and Modules 00-10 of the engineering roadmap.

GitHub full-stack-engineer roadmap
Workflow, Design, Web Fundamentals, HTML5, CSS3, JavaScript, TypeScript, React, Next.js,
Node.js, Express.js, MongoDB

Pasted Markdown source provided in this conversation

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Learning Modules

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Engineering Workflow & Git Collaboration (Company-Grade)

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Every module's project MUST follow this professional workflow.

1. Repository Setup (Fork & Clone)

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Step	Command	Purpose
Fork	GitHub UI -> "Fork"	Create personal copy
Clone	<code>git clone git@github.com:<you>/fullstack-roadmap.git</code>	Copy to local
Add upstream	<code>git remote add upstream git@github.com:org/fullstack-roadmap.git</code>	Track original
Verify	<code>git remote -v</code>	Confirm remotes
Sync main	<code>git fetch upstream && git checkout main && git merge upstream/main</code>	Stay updated

2. Branch Strategy

[GitHub section](#)

Branch	Naming Convention	Purpose
main	main	Production-ready, protected
develop	develop	Integration branch
Feature	feature/JIRA-123-user-auth	New features
Bugfix	bugfix/JIRA-456-login-error	Non-critical fixes
Hotfix	hotfix/JIRA-789-prod-down	Critical production
Release	release/v1.2.0	Release preparation
Chore	chore/update-deps	Maintenance
Docs	docs/api-readme	Documentation
Refactor	refactor/service-layer	Code cleanup
Design	design/figma-login-flow	Design assets

3. Commit Conventions (Conventional Commits)

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Type	Description	Example
feat	New feature	feat(auth): add JWT refresh token
fix	Bug fix	fix(cart): resolve null pointer
docs	Documentation	docs(readme): update setup steps
style	Formatting	style: apply prettier
refactor	Code refactor	refactor(user): extract mapper
perf	Performance	perf(query): add index hint
test	Tests	test(auth): add login tests
build	Build system	build: bump spring boot to 3.3
ci	CI config	ci: add sonar step
chore	Maintenance	chore: update .gitignore
revert	Revert commit	revert: feat(auth) ...
design	Design assets	design: add login wireframes

4. Code Review & Approval

[GitHub section](#)

Step	Actor	Action
1	Author	Open PR, assign reviewers
2	CI	Run build, tests, lint, security scan
3	Reviewer	Review code, comment, request changes
4	Designer	Review UI/UX (if applicable)
5	Author	Address feedback, push fixes
6	Reviewer	Approve
7	Maintainer	Merge

5. Review Rules

[GitHub section](#)

Rule	main	develop
Require PR	Yes	Yes
Required approvals	2	1

Rule	main	develop
Design approval	Yes (UI)	Yes (UI)
Dismiss stale approvals	Yes	Yes
Require status checks	Yes	Yes
Require conversation resolution	Yes	Yes
Require signed commits	Yes	No
Require linear history	Yes	Yes

6. Merge Strategies

[GitHub section](#)

Strategy	When to Use	Command
Squash & Merge	Feature branches (default)	<code>gh pr merge --squash</code>
Rebase & Merge	Linear history preferred	<code>gh pr merge --rebase</code>
Merge Commit	Preserve history	<code>gh pr merge --merge</code>

7. Release & Versioning (SemVer)

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Version	Meaning	Example
MAJOR	Breaking change	<code>1.0.0 -> 2.0.0</code>
MINOR	New feature	<code>1.0.0 -> 1.1.0</code>
PATCH	Bug fix	<code>1.0.0 -> 1.0.1</code>

00 - Design Thinking & Figma

[GitHub section](#)

Sub-Module	Topics
00-01 Design Thinking	Empathize, Define, Ideate, Prototype, Test, Double Diamond, Design Sprint
00-02 UX Research	User Interviews, Surveys, Personas, Empathy Maps, Journey Maps, User Stories
00-03 Information Architecture	Sitemaps, User Flows, Card Sorting, Navigation Patterns
00-04 Wireframing	Low-Fidelity, Mid-Fidelity, High-Fidelity, Sketching, Crazy 8s
00-05 Figma Fundamentals	Frames, Auto Layout, Constraints, Components, Variants, Styles
00-06 Figma Advanced	Design Tokens, Component Sets, Variables, Modes, Prototyping, Smart Animate
00-07 Design Systems	Atomic Design, Tokens, Component Library, Documentation, Storybook
00-08 Prototyping	Clickable Prototypes, Micro-interactions, Transitions, User Testing
00-09 Handoff	Dev Mode, Inspect, Zeplin, Specs, Redlines, Asset Export
00-10 Accessibility Design	Color Contrast (WCAG), Focus States, Touch Targets, Inclusive Design
00-11 Visual Design	Typography, Color Theory, Spacing, Grid Systems, Hierarchy
00-12 Design Critique	Heuristics (Nielsen), Feedback, Iteration, A/B Testing

Project: End-to-End Design of a SaaS Dashboard (Research -> Wireframe -> Hi-Fi -> Prototype -> Handoff)

01 - Internet & Web Fundamentals

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Sub-Module	Topics
01-01	How Internet Works
	Client-Server, DNS, HTTP/HTTPS, HTTP/2, HTTP/3 (QUIC), TCP/IP, Request-Response, Status Codes, CDN basics
01-02	Web Browsers
	Browser Architecture, Rendering Engine, V8, Browser Storage, DevTools, Lighthouse, Core Web Vitals
01-03	Web Security
	HTTPS/TLS, CORS, CSP, XSS, CSRF, SQL Injection, Same-Origin, OWASP Top 10
01-04	Web Protocols
	REST, GraphQL, gRPC, WebSocket, SSE, WebRTC
01-05	Version Control
	Git Internals, Branching, Merge vs Rebase, Cherry-pick, Bisect, Conventional Commits
01-06	Domain & Hosting
	Domain Registrar, DNS Records, SSL/TLS Certificates, CDN, Reverse Proxy

Project: Personal Portfolio with Git Workflow (Fork -> PR -> Review -> Merge)

02 - HTML5

[GitHub section](#)

Sub-Module	Topics
02-01 Fundamentals	Document Structure, Semantic Elements, Metadata, SEO, Open Graph
02-02 Forms	Input Types, Validation, ARIA, File Uploads, FormData API
02-03 Multimedia	Audio, Video, Canvas, SVG, iframe, Picture, Source
02-04 Modern APIs	Web Components, Shadow DOM, Templates, Custom Elements, Web Storage, Drag & Drop
02-05 Accessibility	WCAG 2.2, ARIA Roles, Keyboard Nav, Screen Readers, Focus Management
02-06 PWA Basics	Manifest, Service Workers, Offline, Install Prompt

Project: Accessible Multi-Page Website + PWA

03 - CSS3

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Sub-ModuleTopics	
03-01 Fundamentals	Selectors, Specificity, Box Model, Units, Variables, Cascade, Inheritance
03-02 Layout	Flexbox, Grid, Positioning, Multi-column, Container Queries, Subgrid
03-03 Responsive	Mobile-First, Media Queries, Fluid Typography, Clamp, Container Units
03-04 Modern CSS	Nesting, <code>:has()</code> , <code>:is()</code> , <code>:where()</code> , Layers, Logical Properties, <code>aspect-ratio</code> , <code>accent-color</code>
03-05 Architecture	BEM, Tailwind, CSS Modules, CSS-in-JS, Atomic CSS
03-06 Animation	Transitions, Keyframes, Transform 3D, Scroll-Driven, View Transitions
03-07 Advanced	CSS Houdini, Paint API, Scroll Snap, Custom Properties, <code>color-mix()</code>

Project: Responsive Admin Dashboard with Dark Mode + Animations

04 - JavaScript

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Sub-ModuleTopics	
04-01 Fundamentals	Variables, Data Types, Coercion, Operators, Control Flow, Strict Mode
04-02 Functions	Declarations, Arrow, Closures, IIFE, HOF, Currying, Memoization
04-03 Objects	Creation Patterns, Prototype Chain, Classes, Getters/Setters, Symbol, WeakMap/WeakSet
04-04 Arrays	map/filter/reduce, Destructuring, Spread/Rest, Iterators, Generators
04-05 Modern JS	Template Literals, Modules, Optional Chaining, Nullish Coalescing, Top-Level Await
04-06 Async	Event Loop, Callbacks, Promises, async/await, Combinators, AbortController
04-07 DOM	Selection, Traversal, Events, Delegation, Forms, MutationObserver
04-08 Browser APIs	Fetch, Storage, IndexedDB, Web Workers, IntersectionObserver, ResizeObserver, Geolocation, Clipboard
04-09 Meta Programming	Proxy, Reflect, Symbol, Decorators
04-10 Error Handling	try/catch, Custom Errors, Global Handlers, Error Boundaries
04-11 Memory	Garbage Collection, Memory Leaks, Heap Snapshots

Project: Task Management SPA (Vanilla JS)

05 - TypeScript

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Sub-Module	Topics
05-01 Fundamentals	Type System, Inference, Interfaces vs Types, Enums, Tuples, <code>any</code> vs <code>unknown</code>
05-02 Advanced Types	Union, Intersection, Generics, Conditional, Mapped, Template Literal, Discriminated Unions
05-03 Utility Types	Partial, Required, Pick, Omit, Record, Readonly, Exclude, Extract, ReturnType
05-04 TypeScript + React	Props, Hooks, Events, Context, Generic Components
05-05 Tooling	tsconfig, Strict Mode, Declaration Files, Type Guards, <code>satisfies</code> , <code>const</code> assertions
05-06 Advanced Patterns	Branded Types, Builder Types, Type-Level Programming

Project: Type-Safe API Client Library

06 - React

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Sub-ModuleTopics	
06-01 Fundamentals	Architecture, JSX, Components, Props, State, Lifecycle
06-02 Hooks	useState, useEffect, useContext, useReducer, useRef, useMemo, useCallback, useLayoutEffect, useId, useTransition, useDeferredValue, useSyncExternalStore, Custom Hooks
06-03 Patterns	Compound, Render Props, HOC, Controlled/Uncontrolled, Composition, Slots
06-04 State Management	Local, Context, Redux Toolkit, RTK Query, Zustand, TanStack Query, Jotai
06-05 Routing	React Router v7, Nested, Dynamic, Protected, Data Loaders, Lazy
06-06 Forms	Controlled, React Hook Form, Zod, File Uploads, Multi-step
06-07 API	Fetch, Axios, Error Handling, Cancellation, Optimistic Updates, Infinite Queries
06-08 Performance	React.memo, useMemo/useCallback, Code Splitting, Virtualization, Profiler, Concurrent Features
06-09 Testing	Jest, RTL, Hooks Testing, MSW, Integration Tests
06-10 Production	Error Boundaries, Suspense, RSC, Portals, Build Optimization
06-11 Advanced	Server Components, Streaming, Suspense Boundaries, Transitions
06-12 Animation	Framer Motion, React Spring, Layout Animations

Project: Enterprise React Admin Portal (JWT, RBAC, Charts, Realtime)

07 - Next.js

[GitHub section](#)

Sub-Module	Topics
07-01 Fundamentals	App Router, Layouts, Server vs Client Components, File-Based Routing
07-02 Data	Server Components, Server Actions, Route Handlers, Caching, Revalidation, ISR
07-03 Advanced	Middleware, Dynamic, Parallel, Intercepting Routes, Streaming, PPR
07-04 Auth	NextAuth.js, Sessions, Protected Routes, OAuth
07-05 Optimization	Image, Font, Metadata API, SEO, Edge Runtime

Project: Full-Stack Blog with CMS

08 - Node.js

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Sub-Module	Topics
08-01 Fundamentals	Architecture, V8, Event Loop, Non-Blocking I/O, npm, package.json
08-02 Core Modules	fs, path, os, http, https, events, stream, crypto, buffer, worker_threads, child_process
08-03 Async	Callbacks, Promises, async/await, EventEmitter, Streams (Readable/Writable/Transform)
08-04 APIs	File System, HTTP Server, Env Vars, Process, Clustering, Signal Handling
08-05 Performance	Profiling, Diagnostics, Memory Leaks, perf_hooks

Project: CLI Tool & HTTP Server

09 - Express.js

[GitHub section](#)

Sub-Module	Topics
09-01 Fundamentals	Setup, Routing, Middleware, Error Handling, Template Engines
09-02 REST	Methods, Status Codes, Content Negotiation, API Versioning
09-03 Architecture	MVC, Routes -> Controllers -> Services -> Repositories, DTOs, Validation
09-04 Security	JWT, bcrypt, OAuth2, CORS, Helmet, Rate Limiting, Sanitization
09-05 Advanced	Multer, Nodemailer, Socket.io, Cron, Winston/Pino, Swagger

Project: Production REST API with Auth

10 - MongoDB

[GitHub section](#)

Sub-Module	Topics
10-01 Fundamentals	NoSQL, Documents, Collections, BSON, Shell, CRUD
10-02 Querying	Operators, Projection, Sort, Pagination, Aggregation, Map-Reduce
10-03 Modeling	Embedded vs Referenced, Patterns, Relationships, Denormalization
10-04 Indexing	Types, Compound, Text, Geospatial, TTL, Explain
10-05 Mongoose	Schemas, Models, Validation, Middleware, Population, Virtuals
10-06 Advanced	ACID Transactions, Replica Sets, Sharding, Change Streams, Time Series

Project: E-Commerce Backend (Node + MongoDB)